Can Parametric and Non-Parametric ML Methods Identify and Exploit the Favorite-Longshot Bias in Football Betting Markets?

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Abstract

This study investigates whether the favorite-longshot bias (FLB) in betting markets can be identified and corrected using machine learning. Using 61,974 matches from Football-Data.co.uk spanning 2005 to 2026, we first apply PCA and K-Means clustering to identify four distinct market regimes, finding that FLB is strongly concentrated in matches with heavy home favorites and largely absent elsewhere. We then evaluate three ML models against naive and Shin-adjusted baselines. No model improves raw prediction accuracy over Bet365's implied probabilities, consistent with market efficiency. However, elastic net and XGBoost both achieve statistically well-calibrated probability estimates, correcting the systematic underpricing of favorites that naive normalization fails to address. This improvement is most pronounced in the heavy home-favorite regime, where elastic net reduces the Brier score by 0.342 percentage points relative to the naive baseline. Our results suggest that Machine Learning's contribution in this setting lies in bias correction rather than prediction, and that unsupervised clustering provides a principled way to identify where such corrections matter most.

1 Introduction

1.1 Background

In sports betting, bookmakers set odds for each possible outcome of a match. For example, in a game between Manchester United and Burnley, the odds might be home win (H): 1.30, draw (D): 5.00, and away win (A): 9.50. These odds can be converted into implied probabilities by taking their reciprocals (in this case, the corresponding implied probabilities are 77%, 20% and 10.5%). The total implied probability (107.5%) is greater than 100%, and the 7.5% above 100% is overround, which represents the bookmaker's profit margin. Empirical evidence consistently shows that these implied probabilities are systematically distorted: longshots tend to be overpriced, while favorites tend to be underpriced. This kind of structural pricing bias is widely known as the favorite-longshot bias (FLB).

1.2 Research Questions

RQ1: How does the intensity of the favorite-longshot bias vary across different match profiles? Can unsupervised learning identify distinct market regimes within which this bias differs systematically?

RQ2: Can machine learning models, leveraging pre-match features and Bet365 historical odds, successfully capture bookmaker biases and outperform traditional implied-probability baselines in predicting match outcomes?

1.3 Literature Review

Bookmaker odds are highly informative probability forecasts, but they are systematically distorted by bookmaker margins and behavioural biases. Datasets such as Football-Data.co.uk, which pair odds with match outcomes across major European leagues over two decades, have become a standard testbed for studying these distortions. Constantinou and Fenton (2013) show, using data from 14 European leagues, that market inefficiencies persist even after accounting for the bookmaker margin. Bookmaker odds are therefore a strong but imperfect benchmark.

The most studied distortion is the favourite-longshot bias (FLB): longshots tend to be overpriced and favourites underpriced relative to how often they actually win. Cain, Law and Peel (2003) document this pattern in UK football and link it to bookmaker margin-loading and informed trading. The practical implication is that inverted decimal odds do not give calibrated probabilities directly; they need to be de-vigged first.

Two de-vigging methods are standard. The naive approach simply rescales inverted odds to sum to one, assuming the margin is spread evenly across outcomes. Shin's (1993) model instead assumes a fraction of bets come from informed traders, which produces a heavier margin on longshots. Clarke, Kovalchik and Ingram (2017) compare these methods and find that the naive approach leaves most of the FLB intact, while Shin's adjustment partially removes it. We use both as implied-probability baselines.

Recent work shows that ML models rarely beat bookmaker odds on accuracy or log loss (Wunderlich & Memmert, 2018), because closing odds already absorb most public information. Crucially, accuracy is not the whole story: an ML model that does not beat the market on average can still produce better-calibrated probabilities by correcting systematic biases like FLB. Our study therefore evaluates ML forecasts not only on accuracy, but specifically on whether they reduce favourite-longshot mispricing relative to the naive and Shin baselines, and whether any residual mispricing is exploitable out of sample.

2 Exploratory Data Analysis

2.1 Data Background & Data Preprocessing

Data come from Football-Data.co.uk, covering 68,359 matches across 11 European leagues from July 2005 to April 2026. After removing rows with missing Bet365 1X2 odds or incomplete match statistics, 61,974 matches remain. The target is full-time result (H/D/A), distributed 44.3%, 26.5%, and 29.3%, respectively. We split chronologically: seasons 2005/06–2021/22 for training (45,370 matches), 2022/23–2025/26 for testing (16,604 matches). Figure 1 reports the full breakdown by league division, including match counts, outcome distributions, and train/test splits.

Div	Country	League	n	H %	D %	A %	Train	Test
E0	England	Premier League	7,909	45.6	24.3	30.2	6,460	1,449
SP1	Spain	La Liga	7,890	46.7	25.0	28.3	6,460	1,430
I1	Italy	Serie A	7,875	44.3	26.4	29.2	6,435	1,440
F1	France	Ligue 1	6,827	44.4	26.8	28.9	5,593	1,234
E1	England	Championship	6,548	43.1	26.8	30.1	4,425	2,123
E2	England	League One	6,392	43.3	25.7	30.9	4,271	2,121
D1	Germany	Bundesliga	6,056	44.8	24.7	30.5	4,896	1,160
SP2	Spain	Segunda Division	4,021	44.6	30.3	25.1	2,276	1,745
I2	Italy	Serie B	3,045	40.4	32.2	27.4	1,588	1,457
F2	France	Ligue 2	2,723	41.6	28.8	29.6	1,421	1,302
D2	Germany	2. Bundesliga	2,688	42.9	26.8	30.2	1,545	1,143
Overall			**61,974**	**44.3**	**26.5**	**29.3**	**45,370**	**16,604**

Analysis sample by league division, 2005/06–2025/26. H/D/A shares are percentages of matches; Train covers 2005/06–2021/22, Test covers 2022/23–2025/26.

Figure 1.

2.2 How much "overround" does the betting company take?

Figure 2 shows the distribution of Bet365 overround across the eleven leagues. The overall median margin is approximately 6.0%, but comparing with competitive level, this is more influenced by geography. The three English divisions: E2 (4.8%), E1 (5.0%), E0 (5.3%) post the tightest margins, consistent with Bet365 pricing its home market with sharper odds than overseas competitions. The four continental top tiers (D1, SP1, I1, F1) cluster narrowly between 5.9% and 6.1%, while the four continental second tiers (D2, SP2, I2, F2) carry the widest margins (6.3%–6.6%).

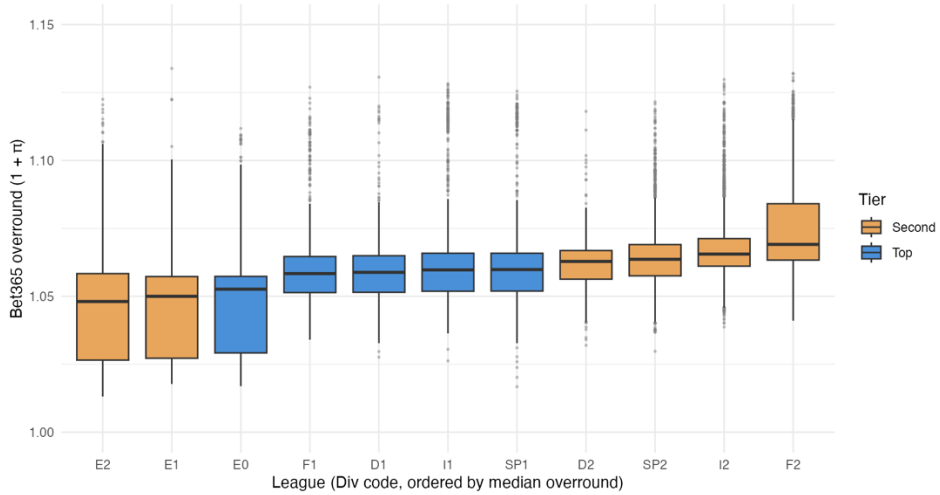


Figure 2. Bet365 overround by league, ordered by median overround. Orange boxes indicate second-tier divisions, blue boxes indicate top-tier divisions.

3 Identifying Market Regimes through Unsupervised Learning

3.1 Feature Construction

We construct 7 features from Bet365's odds alone, excluding post-match statistics, team information, or league labels. After removing the bookmaker's margin to derive de-juiced probabilities, the features fall into three semantically meaningful groups. The first group captures match lopsidedness through four variables (p_fav, p_long, entropy, log_ratio). The second captures market structure via the overround. The third captures outcome skew through the draw probability and home-away asymmetry (q_draw, home_skew). Full definitions are provided in Appendix A.

3.2 Dimensionality Reduction (PCA)

PCA is used to reduce the number of features and remove redundancy. Some variables are correlated, so PCA helps combine them into a smaller set of components while keeping most of the information. Each PC is a weighted combination of these seven features; the weights reveal which features drive each component, and allow us to interpret what each component represents.

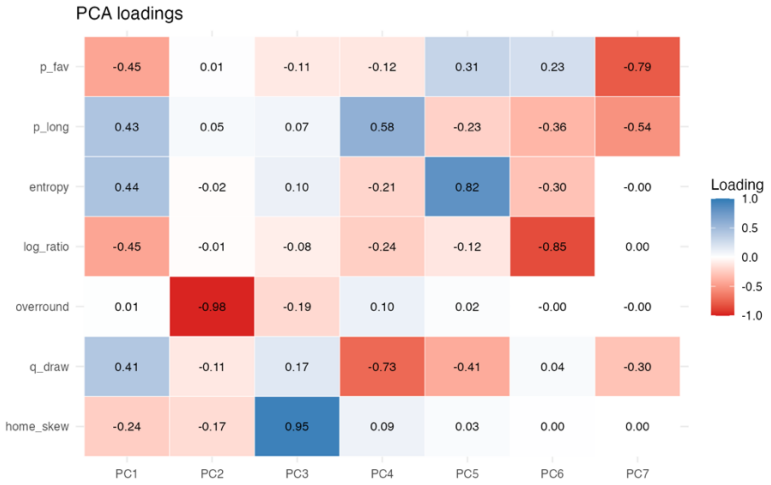


Figure 3. PCA loading matrix for the seven market features across all seven principal components.

PC1 — Match imbalance (70.4% of variance). Five features load on PC1 at comparable magnitudes (|loadings| ≈ 0.41–0.45): p_fav, p_long, entropy, log_ratio, and q_draw. All five describe how lopsided the matchup is, so PC1 functions as a composite imbalance index aggregating their shared signal.
PC2 — Bookmaker margin (14.5% of variance). A single feature overround dominates PC2 with |loading| = 0.98. So PC2 implies almost pure overround.
PC3 — Home-away asymmetry (10.8% of variance). Home_skew dominates PC3 with |loading| = 0.95 while all other features load near zero. PC3 cleanly captures whether the win probability tilts toward the home or away side.

Crucially, the three components recovered by PCA correspond exactly to the three semantic categories proposed in Section 3.1: lopsidedness, market structure, and outcome skew, on independent statistical grounds.

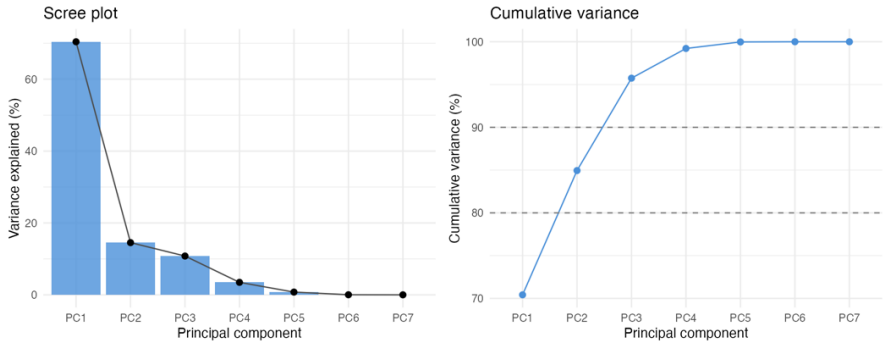


Figure 4. PCA scree plot

3.3 Cluster Identification (K-Means)

After PCA, with each match compressed into three principal components, we group similar matches into market regimes via K-Means. The only open choice is the number of clusters K. We run K-Means $K \in \{2, \ldots, 8\}$with 25 random restarts per K and evaluate three criteria: WSS, Calinski–Harabasz (CH), and silhouette width.

WSS measures how compact the clusters are. CH evaluates both within-cluster cohesion and between-cluster separation, and silhouette width measures how well each observation fits into its assigned cluster.

Silhouette and Calinski–Harabasz both peak at $K = 2$, but this partition merely splits matches along PC1 and provides no analytical value for regime identification. Considering three methods together, K = 4 is the best choice. WSS shows a clear elbow at this point, silhouette improves after K = 3 and reaches a local maximum at K = 4, and the Calinski–Harabasz index stops improving beyond this point.

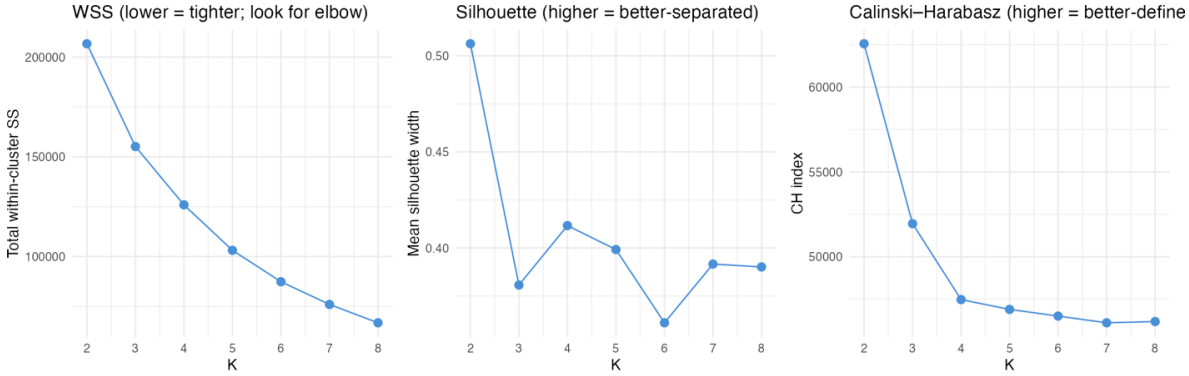


Figure 5.

The four resulting regimes have the following interpretations:

- **Regime 1 (56.4%) — Balanced.** The modal group: three-way contests with high outcome entropy and no home-away tilt.
- **Regime 2 (25.7%) — Moderate home tilt.** Matches where the home side is slightly favoured, with the market pricing the home win at around 55%; the actual home-win rate comes in 1.2pp higher than that.
- **Regime 3 (10.2%) — Extreme home favourite.** The heavy-favourite tail (implied home-win probability above 70%) is concentrated in the Premier League, Serie A, and La Liga, where elite-versus-relegation mismatches are common. The actual home-win rate is 76.1% against an implied 72.7%, a 3.4% gap, and the strongest empirical signal of FLB in our data.
- **Regime 4 (7.7%) — Away tilt.** The reverse case of away-favoured matches, distinguished from the other three along PC3 rather than PC1, realised an away-win rate of 62.3% similarly exceeds the market's 61.2%.

The defining axis across regimes is not lopsidedness alone but its interaction with home–away asymmetry, precisely the structure PC1 and PC3 were designed to separate.

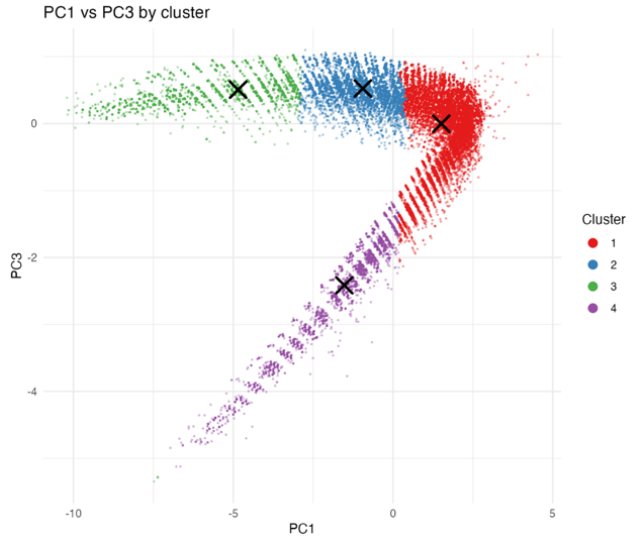


Figure 6.

3.4 Key Findings

With the four regimes established, we now ask the core RQ1 question: Does FLB actually differ across regimes? For each regime, we fit a calibration regression of realised outcome indicators on de-vigged implied probabilities, stacking

H/D/A within each match. A slope of 1 means the bookmaker's prices are perfectly calibrated; a slope above 1 means the market systematically underestimates favourites, which is the signature of FLB. Confidence intervals are clustered at the match level.

Regime	n_{obs}	Slope	95% CI
R1 — Balanced	104,910	1.052	[0.999, 1.105]
R2 — Moderate home tilt	47,817	1.048	[1.014, 1.081]
R3 — Extreme home favourite	18,954	**1.083**	[1.058, 1.107]
R4 — Away tilt	14,241	1.039	[0.995, 1.082]

All four slopes are above 1, suggesting that the FLB exists across the entire market in terms of direction. However, the strength of the bias is not uniform and instead follows a clear gradient. R3, which represents heavy home favourites, shows the strongest distortion with a slope of 1.083 and a confidence interval clearly above 1. R2 also shows a statistically robust bias, with a confidence interval entirely above 1. In contrast, R1 and R4 are only slightly above 1, and their confidence intervals just touch 1, suggesting that in more balanced matches and away-favoured matches, the bias is too weak to be statistically distinguished from a slope of 1. The pattern is also asymmetric: away-favoured matches (R4) exhibit the weakest bias among all groups. Overall, this suggests that the bias is mainly concentrated in matches where the home team is strongly favoured, and largely disappears when the away team is favoured.

Answer to RQ1. Yes, FLB exists, but it doesn't show up everywhere, and it doesn't show up evenly. It gets stronger as the home favourite gets stronger, and is concentrated almost entirely in the heavy-favourite tail (R3). This matters for RQ2: if we evaluate any model on the whole test set lumped together, the strong R3 signal gets diluted by the near-zero R1 and R4 signals, and we'll miss where the model actually helps. We need to look at each regime separately to see where ML adds value.

4 Predictive Modelling of Match Outcomes

4.1 Feature Engineering

Beyond the 7 market features we already have in Section 3.1, we construct 20 additional features describing each team's recent performance. All form features use a fixed rolling window of the last 5 matches, computed strictly from games played before the current kickoff to prevent any look-ahead leakage; matches without 5 complete prior observations for either side are dropped. The features fall into four categories.

- **Attacking efficiency (4 features).** Average goals scored and average shots taken per match over the last 5 games, separately for the home team and the away team.
- **Defensive efficiency (4 features).** Average goals allowed and average shots faced per match over the last 5 games, again split by team.
- **Result tendency (6 features).** Recent results compressed into momentum signals: win rate, draw rate, and points-per-game (W=3, D=1, L=0) over the last 5 matches for each side.
- **Venue-specific performance (6 features).** Form measured only at the relevant venue: the home team's win rate, points-per-game, and goals scored over its last 5 home matches and the away team's win rate, points-per-game, and goals scored over its last 5 away matches. These isolate venue-conditional form, which can diverge sharply from a team's overall form.

The market features capture all the information already reflected in Bet365's odds, so they mainly allow the model to match the market. In contrast, the form features provide additional information, and they are the only way for the model to potentially outperform a simple market-based benchmark.

4.2 Baselines

To evaluate whether ML adds predictive value beyond the information already in bookmaker odds, we establish two benchmark probability forecasts. Any candidate ML model must beat these to justify its complexity.

Baseline 1: Naive de-vigged probability. The simplest benchmark converts Bet365's raw odds to implied probabilities and removes the overround by proportional normalization. For outcome $k \in \{H, D, A\}$:

$$q_k = \frac{1/o_k}{\sum_{j\in\{H,D,A\}} 1/o_j}$$

This requires no fitting and serves as the most transparent reference point: a forecaster with full access to Bet365's pricing but no attempt to correct known biases.

Baseline 2: Shin-adjusted probability. Proportional normalization assumes the overround is distributed evenly across outcomes, which is an assumption the FLB literature has long shown to be wrong. Bookmakers systematically shade longshot odds to protect against informed bettors, inflating longshot implied probabilities and deflating favourite ones. Shin (1993) derives the classical closed-form correction under the assumption that a fraction z of wagers comes from insiders. Given raw implied quotes $\pi_k = 1/o_k$ with overround sum $B = \sum_k \pi_k$:

$$p_k = \frac{\sqrt{z^2 + 4(1-z)\pi_k^2/B} - z}{2(1-z)}, \text{subject to } \sum_k p_k = 1$$

The insider fraction $z \in (0, 0.5)$ is recovered per match via root-finding on the normalization constraint.

Test-set performance. Both baselines are evaluated on the held-out 2022/23–2025/26 test set (n = 16,333). Shin marginally improves on naïve-Brier 0.6025 vs 0.6028 and log-loss 1.0065 vs 1.0071 and substantially improves calibration (slope 1.037 vs 1.082, with the Shin CI's lower bound at 1.002 versus naive at 1.046). Both, however, remain statistically miscalibrated: neither 95% CI contains 1.0. Section 4.3 reports the consolidated comparison against the ML candidates.

4.3 Parametric Models

We train an elastic net, which is a logistic regression with built-in feature shrinkage on 27 features and compare it against two baselines.

Two hyperparameters (α mixing L1/L2 and overall strength λ) are chosen by expanding-window time-series cross-validation. Training folds always precede their validation folds in calendar time, preventing future information from leaking into past predictions. The CV-optimal configuration is $\alpha = 1.0$ (pure Lasso) at $\lambda = 0.0028$.

Model	Brier	Log-loss	Calib. slope	95% CI
Naive de-vigged	0.6028	1.0071	1.082	[1.046, 1.119]
Shin	0.6025	1.0065	1.037	[1.002, 1.072]
Elastic net	**0.6024**	**1.0065**	**1.015**	**[0.980, 1.049]**

Test set: 16,333 matches, seasons 2022/23–2025/26.

The results tell two slightly different stories. In terms of predictive accuracy, all three models perform almost the same. The tiny 0.0001 difference in Brier score between the elastic net and Shin model is basically just noise. This suggests that Bet365's opening odds already capture most of the useful predictive information. As a result, the 20 form features we added do not really improve performance once the 7 market features are included. In this sense, the data provide support for the efficient market hypothesis.

On calibration the picture reverses. Naive de-vigging produces slope 1.082, CI entirely above 1.0, the FLB documented in part 3.4 at the global level. Shin partially corrects it (slope 1.037, CI lower bound 1.002 still excludes 1.0). The elastic net produces a slope of 1.015 with CI [0.980, 1.049], which is the only model whose 95% CI contains 1.0, statistically indistinguishable from a well-calibrated forecaster.

What the model is actually doing. Bet365 is already quite good at predicting outcomes, but it has a consistent bias. It usually gets the stronger team right, but tends to slightly underestimate strong favourites. For example, when it gives Manchester City a 72.5% chance of winning, the actual win rate is closer to 76%, which is the FLB. The elastic net does not really bring in new information, it just adjusts these probabilities by pushing favourites up a bit and longshots down. So the overall accuracy stays about the same, but the probabilities become more realistic. We can also see this from the Lasso results, where form features get almost zero weight and market features stay important but are reweighted. In the end, the improvement comes from correcting the bias in the odds rather than adding anything new.

4.4 Non-Parametric Models

The elastic net showed that linearly re-weighting market features can fix FLB without improving prediction. To test whether non-linear interactions among the features carry additional signal, we fit two tree-based models with different inductive biases. Both are tuned by 5-fold time-series CV.

4.4.1 Random Forest

Cross-validation chose a deliberately conservative configuration: 500 trees, where each tree only considers 3 features at every split (out of the 27 available) and is forced to stop splitting whenever a branch contains fewer than 50 matches. This keeps the trees shallow and prevents them from carving the data into thin parts. The fact that CV pulled toward such cautious settings is itself a signal that the data don't reward complex models.

Model	Brier	Log-loss	Calib. slope	95% CI
Naive de-vigged	0.6028	1.0071	1.082	[1.046, 1.119]
Shin	0.6025	1.0065	1.037	[1.002, 1.072]
Elastic net	0.6024	1.0065	1.015	[0.980, 1.049]
Random Forest	**0.6065**	**1.0130**	**1.050**	**[1.014, 1.087]**

So we can see that RF is the worst predictor in table, worse than every baseline, including the naive one. Brier 0.6065 vs naive 0.6028 means a tuned random forest with 20 extra team-form features produces *less accurate* probability forecasts than a one-line formula that just inverts Bet365's odds. RF also fails to fix FLB. Slope 1.050 with CI [1.014, 1.087] excludes 1.0 statistically not well-calibrated, and worse than even Shin's analytical correction (1.037).

Random forests tend to shrink extreme probabilities toward the mean because they average predictions across many trees. Each tree is usually based on a limited sample of the data, so it tends to make more conservative predictions. When these are averaged together, strong favourites end up being assigned lower probabilities than they should be. As a result, the model systematically underestimates the chances of heavy favourites.

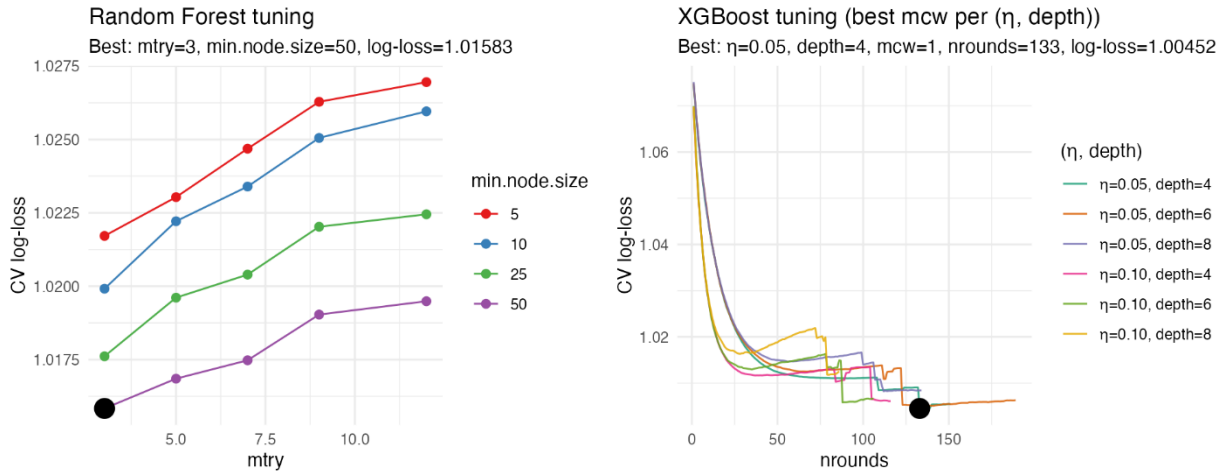


Figure 7.

4.4.2 XGBoost

XGBoost is also tree-based, but it builds the trees in a different way. Instead of training many independent trees and averaging them like RF, XGBoost grows trees one at a time sequentially, where each new tree is trained specifically to fix the mistakes the previous trees made. The final prediction is the cumulative output of all trees combined. We use XGBoost rather than classical AdaBoost because XGBoost optimizes log-loss directly, which aligns with our calibration-based evaluation, whereas AdaBoost's exponential loss is designed for hard classification and tends to produce poorly calibrated probability estimates.

Cross-validation chose a learning rate of 0.05 (each tree contributes only a small correction so the model takes small, careful steps), a maximum tree depth of 4 (each tree only splits the data four times before stopping), and stopped at 133 trees (after which adding more trees stopped helping on the validation set). Same shallow-tree story as RF, but XGB compensates by stacking 133 of these small trees in sequence, each one cleaning up after the last.

Model	Brier	Log-loss	Calib. slope	95% CI
Naive de-vigged	0.6028	1.0071	1.082	[1.046, 1.119]

Model	Brier	Log-loss	Calib. slope	95% CI
Shin	0.6025	1.0065	1.037	[1.002, 1.072]
Elastic net	0.6024	1.0065	1.015	[0.980, 1.049]
Random Forest	0.6065	1.0130	1.050	[1.014, 1.087]
XGBoost	**0.6029**	**1.0073**	**1.022**	**[0.988, 1.056]**

On prediction, XGB ties with everything else. Brier 0.6029 versus naive 0.6028 is a 0.0001 gap, pure noise, which means that the market is sharp, ML does not beat it. On calibration, XGB joins the elastic net as the second model whose 95% CI contains 1.0, statistically indistinguishable from a well-calibrated forecaster. XGB fixes FLB.

The contrast with RF is the whole point of this section. XGB optimizes log-loss directly: every new tree is trained to reduce the prediction error of the previous trees. So when the training data contains heavy favourites that previous trees underestimated, the next tree's only job is to push those probabilities up because that's exactly what reduces the loss. RF has no such feedback loop: every tree is grown independently, and the final averaging step can only pull probabilities toward the center, never push them outward. **RF averages, XGB learns from mistakes.** That's why one fixes FLB and the other reproduces it.

4.5 Per-Regime Evaluation

Part 4.3 and Part 4.4 found that, on the full test set, no model meaningfully beats Bet365 on prediction accuracy, but elastic net and XGBoost both eliminate FLB on calibration. These are *averaged* numbers, however, pooled across all four regimes. If ML's advantage is concentrated in one specific regime, the pooled average dilutes that advantage. This section splits the test set by regime and evaluates each model separately.

Regime	*Model*	*Brier*	*Lift vs. Naive*
1 Balanced *n = 9,393*	**Elastic Net**	**0.6501**	**+0.024**
	Shin (1993)	0.6502	**+0.008**
	Naive baseline	0.6503	—
	XGBoost	0.6505	**-0.021**
	Random Forest	0.6530	**-0.268**
2 Moderate Home Tilt *n = 4,473*	**Elastic Net**	**0.5859**	**+0.041**
	XGBoost	0.5859	**+0.036**
	Shin (1993)	0.5860	**+0.027**
	Naive baseline	0.5863	—
	Random Forest	0.5893	**-0.304**
3 Extreme Home Favourite *n = 1,311*	**Elastic Net**	**0.3743**	**+0.342**
	Shin (1993)	0.3756	**+0.203**
	XGBoost	0.3758	**+0.183**
	Naive baseline	0.3777	—
	Random Forest	0.3841	**-0.641**
4 Away Tilt *n = 1,156*	**Shin (1993)**	**0.5352**	**+0.047**
	Naive baseline	0.5357	—
	Elastic Net	0.5369	**-0.123**
	XGBoost	0.5397	**-0.399**
	Random Forest	0.5476	**-1.191**

Table 4.5. Per-regime lowest Brier scores. Rows are sorted within each regime from best to worst. Highlighted row = best model by Brier score. Lift reported in percentage points vs. the naive de-vigged baseline.

Figure 8. Figure 8 reports Brier scores within each regime for all five models, with the best model in each regime highlighted. The regime conditional picture is sharply different from the pooled one.

R3 (Extreme Home Favourite) is the only regime where ML clearly wins and it wins by a lot. Elastic net beats the naive baseline by 0.342 Brier percentage points in R3, with XGBoost gaining 0.193. By comparison, the lift in R1 and R2 is only 0.024–0.041. FLB is strongest where R3 lives, and ML extracts the most value where R3 lives. These two findings reinforce each other.

R1 (Balanced) is where ML offers almost nothing. R1 covers 56% of the test set; the top three models cluster within 0.0003 Brier of each other, which is statistical noise. The market is itself well-calibrated here, so there is no FLB for ML to learn from.

The result in R4 (Away tilt) looks different, but this can be explained by data limitations rather than a contradiction. The sample size is relatively small, and the calibration pattern for away-favoured matches may have shifted between the training and test periods. As a result, more flexible models like elastic net may overfit to patterns that no longer hold. This implies that machine learning only improves calibration when there is a stable bias to learn and enough data to estimate it reliably.

Overall, the results show that machine learning does not improve predictive accuracy, as Bet365's odds already capture most of the available and valuable information. However, ML methods can still improve the quality of the probabilities by correcting the systematic bias in the odds. This improvement is not uniform, but mainly occurs in the R3 regime, where the FLB is strongest. This implies that the clustering structure is not just for presentation, but actually helps identify where probability adjustments matter most, even if it does not lead to more correct predictions.

5 Conclusion

This project asked whether the FLB is still detectable and correctable in modern European football betting markets.

RQ1: Yes, the bias is real and concentrated. Unsupervised clustering on seven market-derived features finds four interpretable match regimes, and FLB is sharply concentrated in just Regime 3, the heavy home-favourite tail. The other three regimes show only mild miscalibration, and the away-favoured regime is essentially well-calibrated. Three independent diagnostics: PCA loadings, K-Means centroids in PC space, and per-regime calibration regressions converge on the same partition result.

RQ2: ML cannot improve prediction, but it can correct calibration where the bias exists. At the aggregate level, none of the ML models really beat Bet365 in terms of Brier score or log-loss. All five probability estimates are very close, within a 0.004 Brier range, which shows that the bookmaker's opening odds already capture most of the useful information. The 20 team-form features we added do not seem to provide any extra signal beyond the seven market features, which is consistent with Lasso shrinking them to zero and their low importance in Random Forest and XGBoost. What ML can still do is adjust the existing probabilities to correct the FLB. In particular, Elastic Net and XGBoost are the only models with calibration slopes that are not statistically different from 1.0. When we break things down by regime, this improvement is mainly concentrated in R3, where Elastic Net achieves a 0.342 percentage point improvement in Brier score over the naive baseline, much larger than in R1 or R2, and exactly where we previously identified the strongest FLB.

Limitations.
Several limitations should be kept in mind when interpreting our results.

- **Form features ignore opponent quality and season boundaries.** Our 20 team-form features use raw 5-game rolling windows without adjusting for the strength of recent opponents, and without restricting the window to the current season. A team scoring freely against relegation-tier defences carries little signal about an upcoming fixture against an elite side, and early-season matches inherit features from the previous season, particularly problematic for promoted and relegated teams. The fact that Lasso shrinks all 20 form features to zero is consistent with this: under our specification, they carry no marginal signal beyond what is already encoded in Bet365's odds.
- **Limited statistical power in R3.** The Extreme Home-Favourite regime, where ML's calibration gains are concentrated, contains only 1,311 test-set matches. Confidence intervals around the per-regime Brier and slope estimates remain correspondingly wide, and the 0.342-percentage-point improvement should be read as suggestive rather than definitive.

Appendix

Appendix A. Feature Definitions

Match lopsidedness (how unbalanced the matchup is)

- **p_fav**: the market's stated confidence that the stronger side will win
- **p_long**: the chance the weaker side wins the match
- **entropy**: a single-number summary of overall outcome uncertainty, high when all three results look plausible, low when one outcome clearly dominates
- **log_ratio**: how lopsided the matchup is on a logarithmic scale, designed to remain sensitive at the extremes where linear differences flatten out

Market structure (how the bookmaker prices the match)

- **overround**: the profit margin Bet365 builds into the prices

Outcome skew (the shape of the three-way distribution)

- **q_draw**: the implied probability of a tie, which shifts systematically with each league's scoring environment (low-scoring divisions produce more draws)
- **home_skew**: how much the match is tilted toward the home team compared to the away team

Coding Chunks

3.1 Feature Engineering

```
data <- data |>
  mutate(
    # 1. Implied probabilities from B365 decimal odds
    r_H           = 1 / B365H,
    r_D           = 1 / B365D,
    r_A           = 1 / B365A,
    overround_raw = r_H + r_D + r_A - 1,

    # 2. Normalized probabilities (basic overround removal)
    q_H = r_H / (1 + overround_raw),
    q_D = r_D / (1 + overround_raw),
    q_A = r_A / (1 + overround_raw),

    # 3. Favourite / longshot decomposition
    p_fav     = pmax(q_H, q_A),
    p_long    = pmin(q_H, q_A),
    log_ratio = log(p_fav / p_long),

    # 4. Auxiliary regime features
    entropy   = -(q_H * log(q_H) + q_D * log(q_D) + q_A * log(q_A)),
    q_draw    = q_D,
    home_skew = q_H - q_A
  )
```

4.1 Construction of past-only 5-match form features

```
roll5 <- function(x) {
  zoo::rollapplyr(dplyr::lag(x, 1), width = 5, FUN = mean,
                  fill = NA, partial = FALSE)}
all_r <- long |>
  arrange(team, Date, match_id) |>     # chronological order within each team
  group_by(team) |>                    # apply the roll WITHIN a team's history
  mutate(
```

```
    r_gf = roll5(goals_for),           # last-5 mean goals scored
    r_ga = roll5(goals_against),       # last-5 mean goals conceded
    r_wr = roll5(as.numeric(is_win)),  # last-5 win rate
    r_pp = roll5(points))              # last-5 points-per-game
```

4.3 Expanding-window time-series cross-validation

```
fold_tr <- ymd(c("2017-08-01","2018-08-01","2019-08-01","2020-08-01","2021-08-01"))
fold_va <- ymd(c("2018-08-01","2019-08-01","2020-08-01","2021-08-01","2022-08-01"))

for (i in 1:5) {
  tr <- which(train$Date <  fold_tr[i])                            # past only
  va <- which(train$Date >= fold_tr[i] & train$Date < fold_va[i])  # next season

  fit <- glmnet(X_train[tr, ], y_train[tr],
                family = "multinomial", alpha = a, lambda = lams,
                type.multinomial = "grouped")
  # ... compute multinomial deviance on (X_train[va,], y_train[va])
}
```

Reference

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